

Plugging into the Future: An Exploration of Electricity Consumption Patterns

TEAM ID	SWTID-2026-4030
PROJECT TITLE	Plugging into the Future: An Exploration of Electricity Consumption Patterns
MAX MARKS	3 MARKS

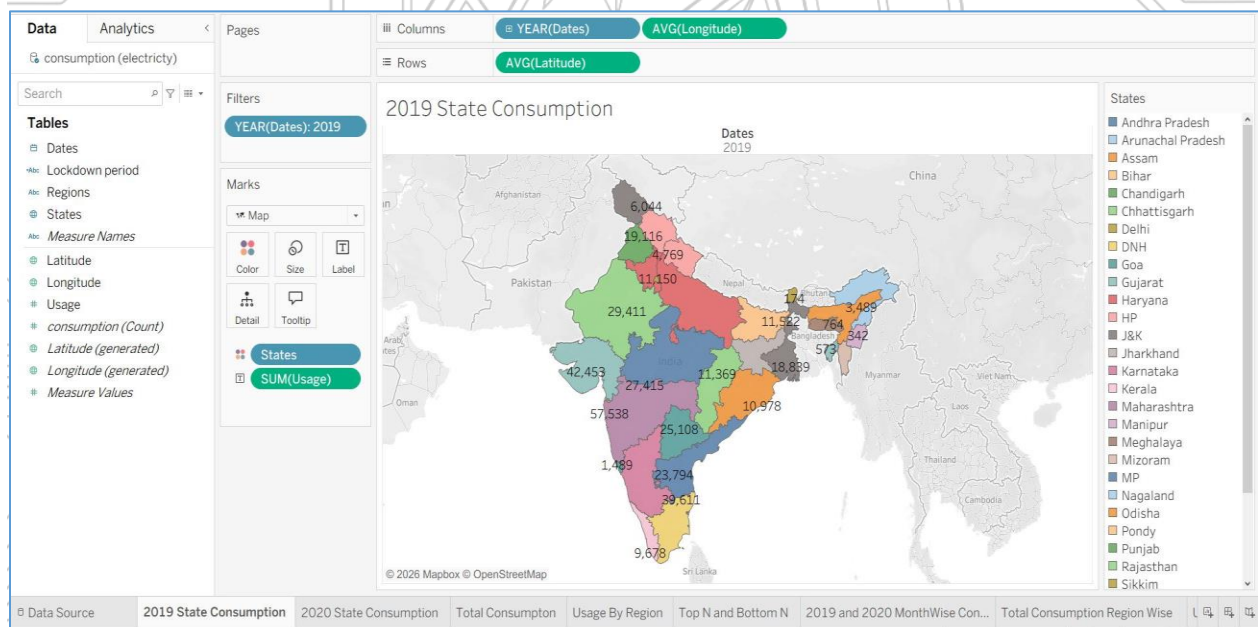
6.3 – Data Preparation

CHART NAME	CHART TYPE
2019 State Consumption	Map Chart
2020 State Consumption	Map Chart
Total Consumption	Bar Chart (Column Type)
Usage By Region	Side-by-side bar Chart
Top N and Bottom N	Bar Chart
2019 and 2020 Monthwise Consumption	Histogram
Total Consumption Region Wise	Pie Chart
Usage Before and After Lockdown	Side-by-side Column Chart
Region Wise State Usage	Horizontal Bar Chart
Quarter Usage	Bar Chart
Metro City State Usage	Symbol Map
Usage by year	Line Chart

SHEET-1: 2019 State Consumption

1. Open Tableau Connect to the Data Source click on file and then new.
2. Then create a sheet.
3. Name it as 2019 State Consumption.
4. Drag and drop Year (Dates) and Avg(longitude) to Columns Shelf.
5. Drag and drop Avg (Latitude) to Rows Shelf.
6. Click on Show me in toolbar and select Map.
7. In filters add Year(dates) as 2019.
8. On Marks Card , Drag and drop States to Color.
9. To display the values click the show mark labels on the label of the marks card.

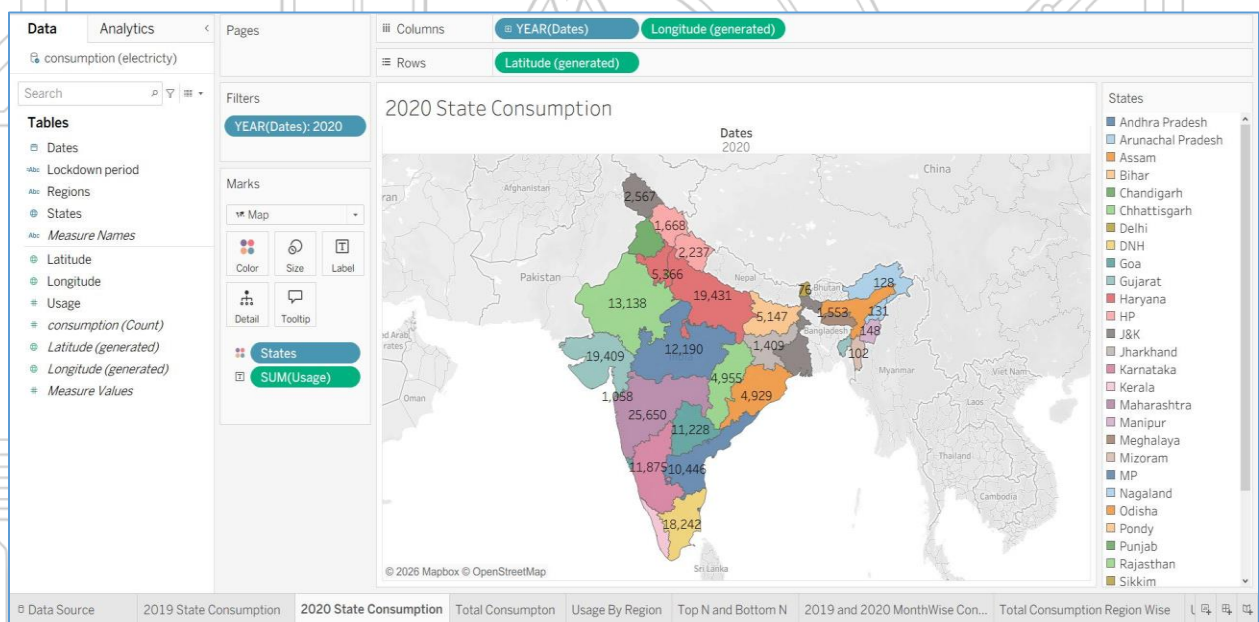
OUTPUT:



SHEET-2: 2020 State Consumption

1. Click on new sheet and name it as 2020 State Consumption.
2. Drag and drop Year (Dates) and Avg(longitude) to columns.
3. Drag and drop Avg (Latitude) to rows.
4. Click on Show me in toolbar and select Map.
5. In filters add Year(dates) as 2020.
6. On Marks Card, Drag and drop States to Color.
7. To display the values click the show mark labels on the label of the marks card.

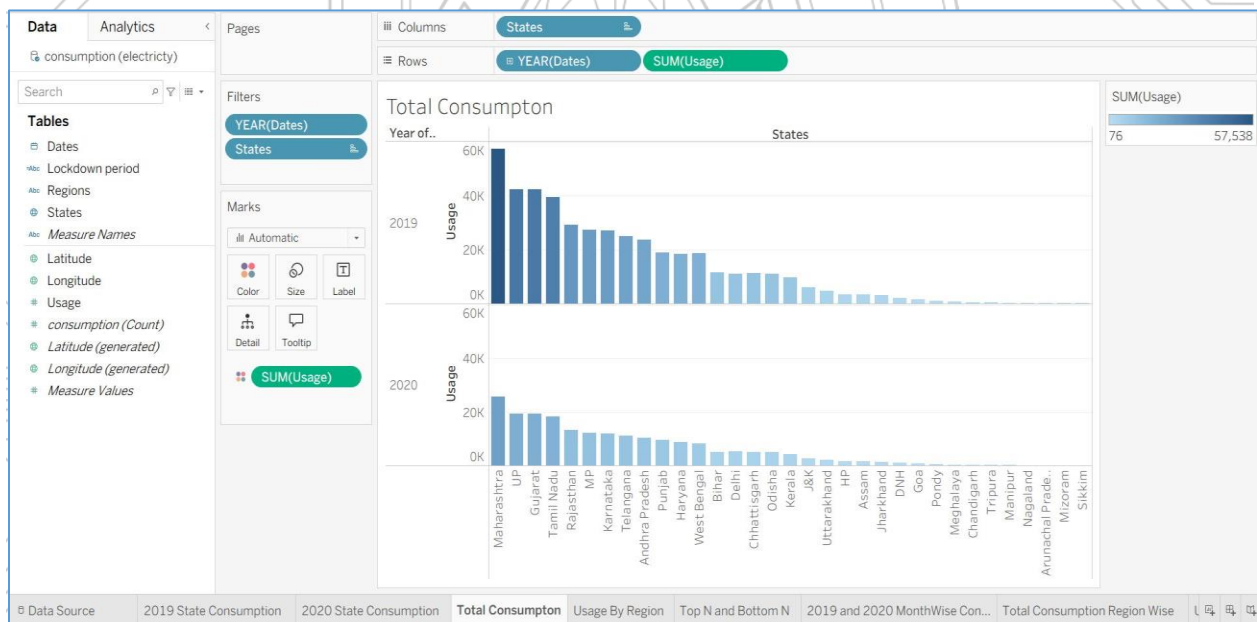
OUTPUT:



SHEET-3: Total Consumption

1. Click on new sheet and name it as Total Consumption.
2. Drag and drop States to columns.
3. Drag and drop Year (Dates) and Sum(usage) to rows.
4. In filters add Year(dates) and select 2019 and 2020, add States also.
5. On Marks Card, Drag and drop SUM(Usage) to Color.
6. Click on the sorted on the toolbar.
7. Click Fit menu (toolbar).
8. Select Entire View.

OUTPUT:

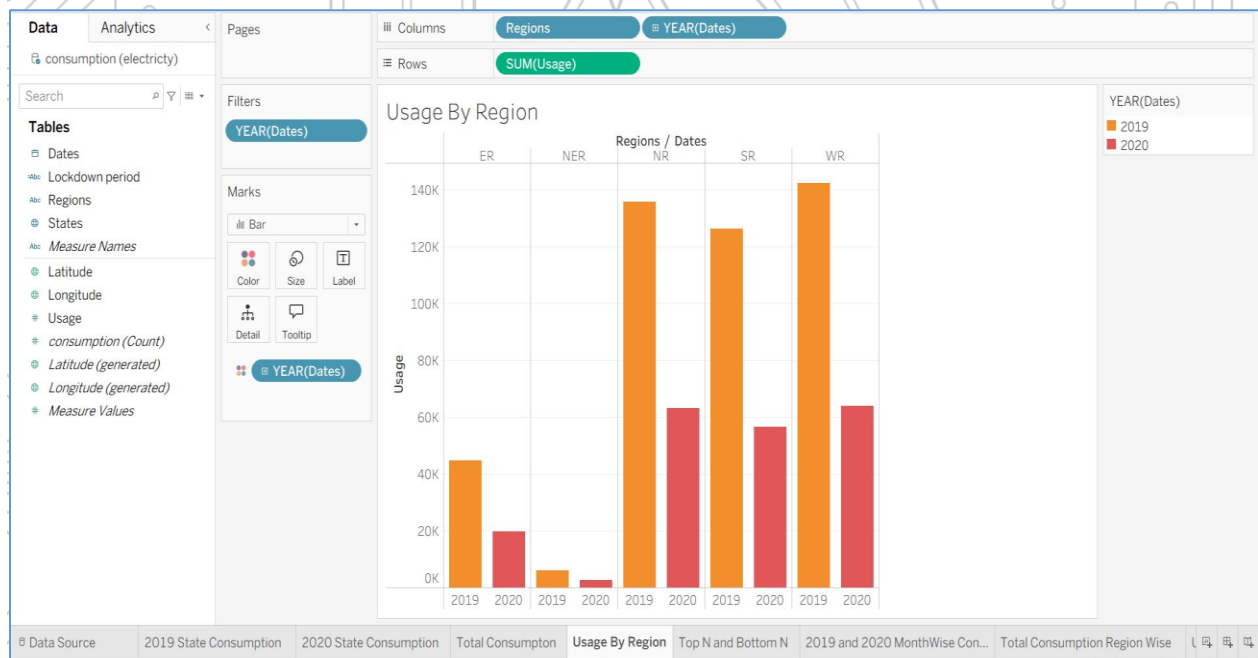


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SHEET-4: Usage By Region

1. Click on new sheet and name it as Usage By Region.
2. Drag and drop Regions and Year (Dates) to columns.
3. Drag and drop Sum(usage) to rows.
4. Click on Showme in toolbar and select Bar.
5. In filters add Year(dates) and select 2019 and 2020.
6. On Marks Card, Drag and drop YEAR(Dates) to color.
7. Drag SUM(Usage) to Label.

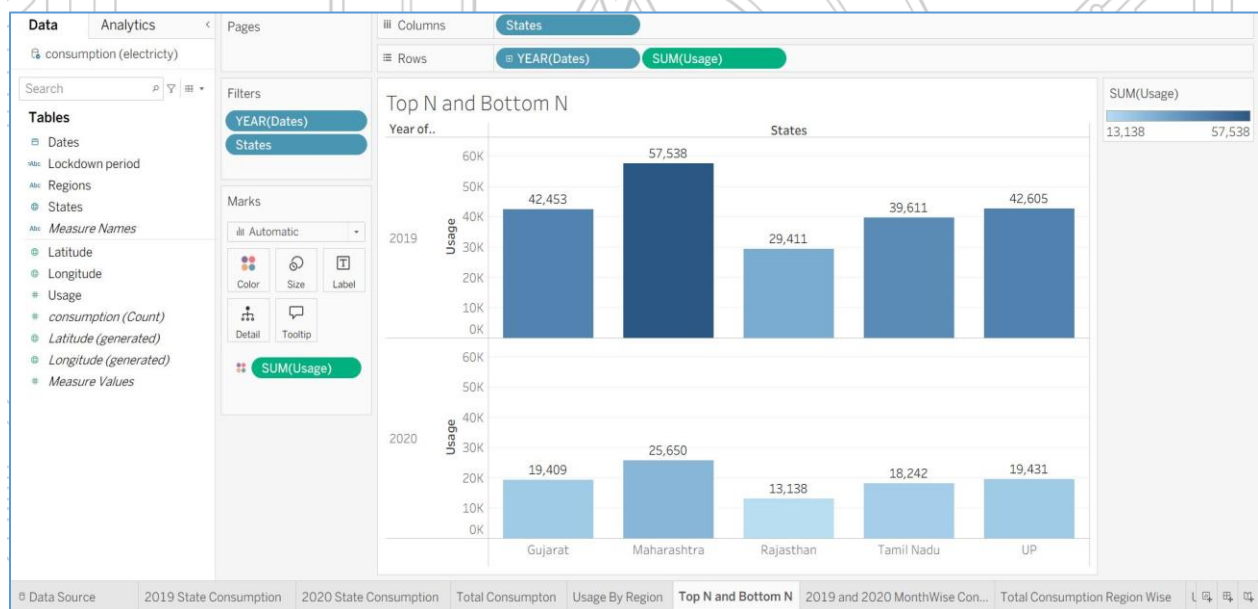
OUTPUT:



SHEET- 5: Top N and Bottom N

1. Click on new sheet and name it as Top N and Bottom N.
2. Drag and drop States to columns.
3. Drag and drop YEAR(Dates) and Sum(usage) to rows.
4. Click on Showme in toolbar and select Automatic.
5. In filters add Year(dates) and select 2019 and 2020 and select states as Gujarat, Maharashtra, Rajasthan, Tamil Nadu and UP.
6. On Marks Card, Drag and drop sum (usage) on the color of the marks card.

OUTPUT:



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SHEET-6: 2019 and 2020 Monthwise Consumption

1. Click on new sheet and name it as 2019 and 2020 Monthwise Consumption.
2. Drag and drop Year (Dates), to columns.
3. Drag and drop Quarter(dates) to columns.
4. Drag and drop Month(Dates) to columns.
5. Drag and drop Sum(usage) to rows.
6. Click on Showme in toolbar and select Bar.
7. In filters Section drag and drop Year(dates) and select 2019 and 2020.
8. On Marks Card, Drag and drop MONTH(Dates) to color.

OUTPUT:

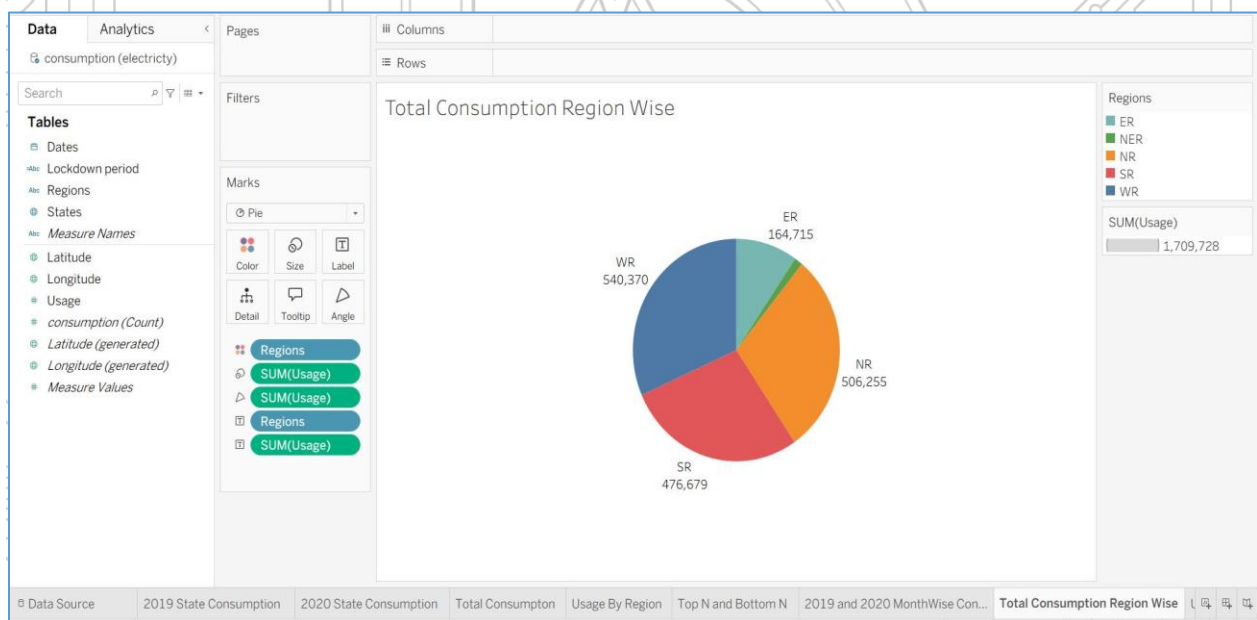


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SHEET-7: Total Consumption Region Wise

1. Click on new sheet and name it as Total Consumption Region Wise.
2. Click on Showme in toolbar and select Pie.
3. On Marks Card, Drag and drop Regions to color.
4. On Marks Card, Drag and drop Sum(usage) to Size.
5. On Marks Card, Drag and drop Sum(usage) to angle.
6. On Marks Card, Drag and drop Regions to Label
7. On Marks Card, Drag and drop sum(usage) to Label.

OUTPUT:



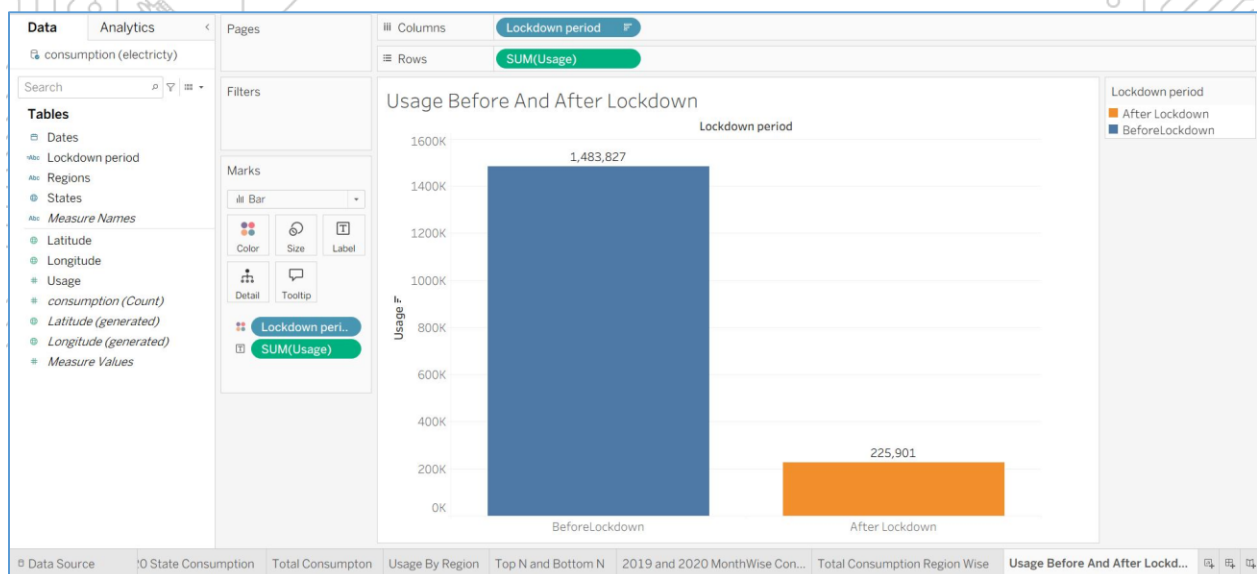
SHEET-8: Usage Before and After Lockdown

1. Click on new sheet and name it as Usage Before and After Lockdown.
2. Click Analysis → Create Calculated Field.
3. Name: Lockdown period.
4. Formula: If MONTH([Dates])<=3 THEN
"After Lockdown"

ELSE
"BeforeLockdown"

END.
5. Click OK.
6. Drag and drop the Lockdown period on the columns Shelf.
7. Drag and drop the SUM(Usage) on the Rows Shelf.
8. On the Marks Card, click the drop-down menu.
9. Select Bar.
10. Drag and drop the Lockdown period to color on the Marks Card.
11. Drag and drop the SUM(Usage) to the label on the Marks card.

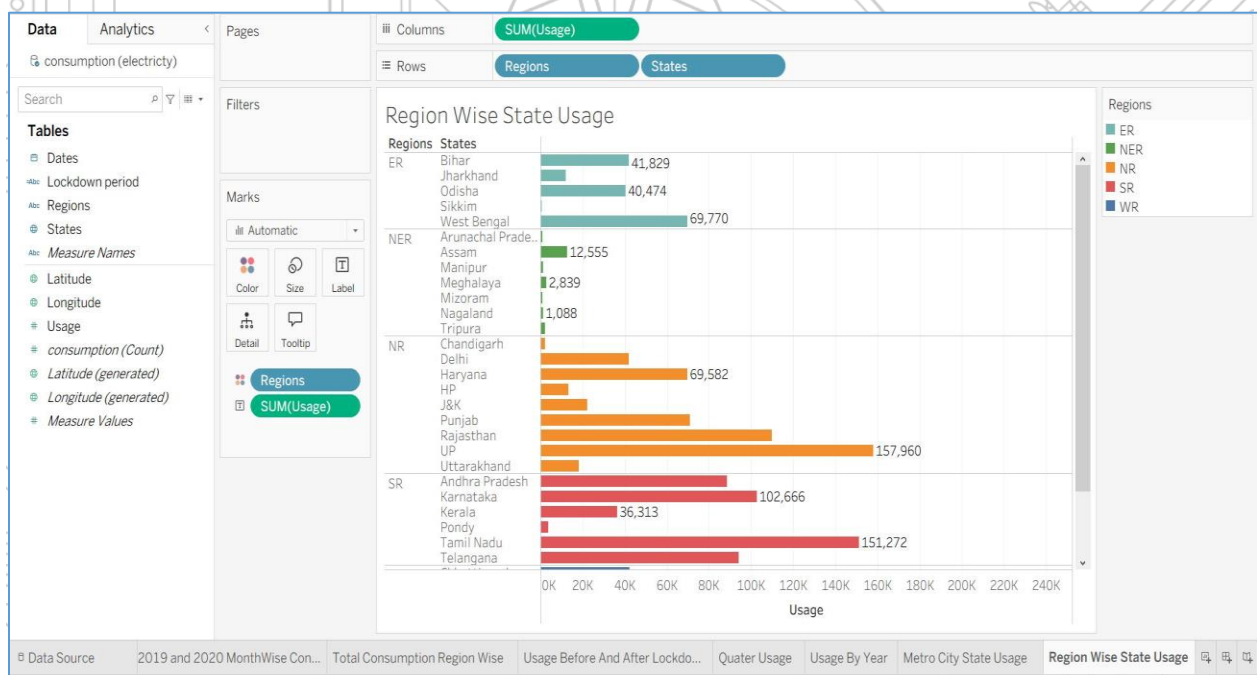
OUTPUT:



SHEET-9: Region Wise State Usage

1. Click on new sheet and name it as Region Wise State Usage.
2. Drag and drop SUM(Usage) in the Columns Shelf.
3. Drag and drop Regions in the Rows Shelf.
4. Drag and drop states in the Rows Shelf.
5. Drag and drop the Regions to color on the Marks Card.
6. Drag and drop the SUM(Usage) to the label on the Marks card.

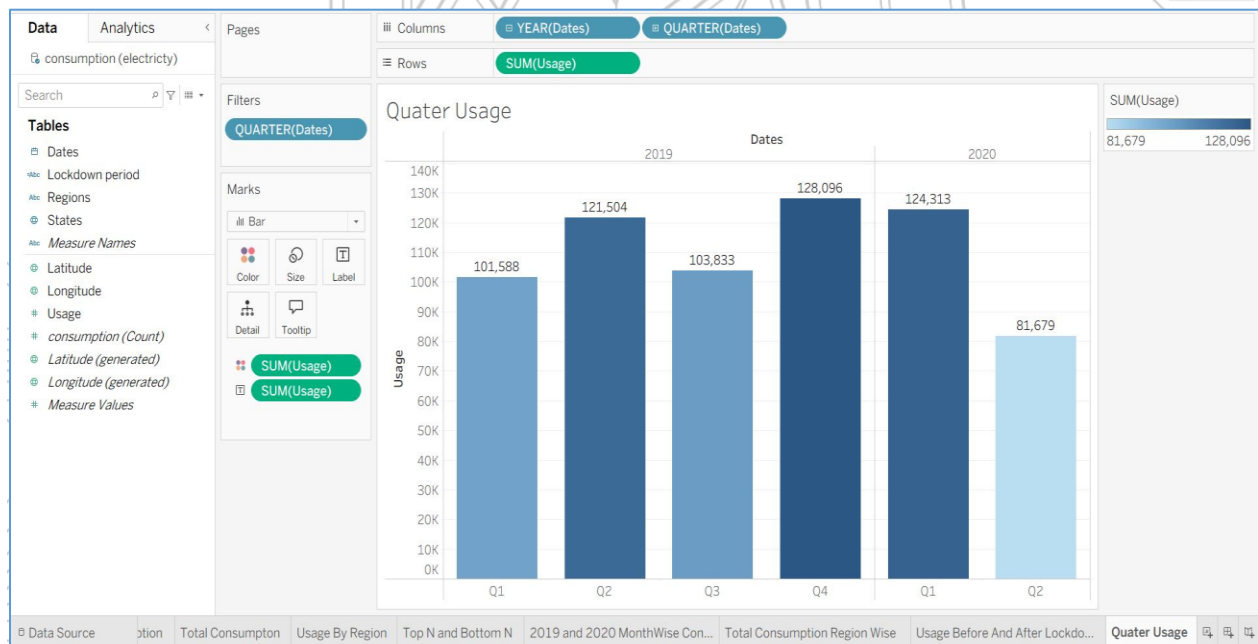
OUTPUT:



SHEET-10: Quarter Usage

1. Click on new sheet and name it as Quarter Usage.
2. Drag and drop YEAR(Dates) to Columns Shelf.
3. Drag and drop Quarter(dates) to Columns Shelf.
4. Drag and drop Sum(usage) to Rows Shelf.
5. In filters Section drag and drop Year(dates) and select Quarter (Dates).
6. On the Marks Card, click the drop-down menu.
7. Select Bar.
8. On Marks Card, Drag and drop Sum(usage) to color.
9. On Marks Card, Drag and drop Sum(usage) to label.

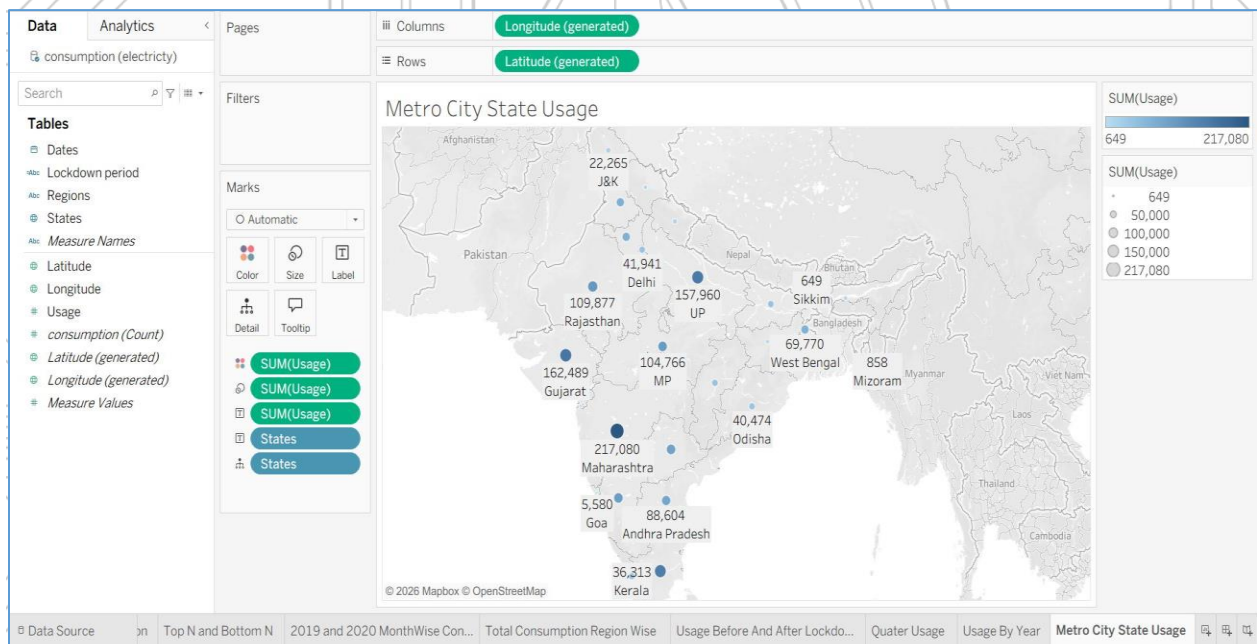
OUTPUT:



SHEET-11: Metro City State Usage

1. Click on new sheet and name it as Metro City State Usage.
2. Drag and drop Longitude (generated) to Columns Shelf.
3. Drag and drop Latitude (generated) to Columns Shelf.
4. On Marks Card, Drag and drop Sum(usage) to color.
5. On Marks Card, Drag and drop Sum(usage) to size.
6. On Marks Card, Drag and drop Sum(usage) to label.
7. On Marks Card, Drag and drop States to label.
8. On Marks Card, Drag and drop States to detail.

OUTPUT:



SHEET-12: Usage by year

1. Click on new sheet and name it as Usage by year.
2. Drag and drop YEAR(Dates) to Columns Shelf.
3. Drag and drop MONTH(Dates) to Columns Shelf.
4. Drag and drop SUM(Usage) to Rows Shelf.
5. On Marks Card, Drag and drop YEAR(Dates) to color.
6. On Marks Card, Drag and drop SUM(Usage) to label.
7. On the Marks Card, click the drop-down menu.
8. Select Automatic.

OUTPUT:

